CAUTION

KEEP OUT OF REACH OF CHILDREN
READ SAFETY DIRECTIONS BEFORE OPENING OR USING

Chess®

Insecticide

syngenta®

ACTIVE CONSTITUENT: 500 g/kg PYMETROZINE

GROUP	9B	INSECTICIDE

For control of Aphids and suppression of Whiteflies in certain vegetable crops, Cotton, Cut Flowers, Nursery stock (non-food), Stonefruit, Almonds and Pistachios as per the Direction for Use.

1, 5 and 20 kg

Syngenta Australia Pty Ltd
Level 1, 2-4 Lyonpark Road, Macquarie Park NSW 2113

In a transport emergency dial 000, Police or Fire Brigade.
For specialist advice in an emergency only, call 1800 033 111 (24 hours).

APVMA Approval No.: 53311/63371

TM

DIRECTIONS FOR USE

RESTRAINTS
DO NOT apply more than 2 applications of CHESS per crop.
DO NOT apply consecutive applications of CHESS or other Group 9B insecticides within and between seasons. Use of CHESS is subject to Crop Life Resistance Management strategies for green peach aphid and silverleaf whitefly. In cotton, use CHESS in accordance with the Cotton Industry's Insecticide Resistance Management Strategy

Crop	Pest	Rate	WHP	Critical Comments
Brassica vegetables (including Broccoli, Brussels sprouts, Cabbage, Cauliflowers)	Cabbage Aphid (*Brevicoryne brassicae),* Green Peach Aphid (*Myzus persicae*)	200 g/ha	7 days (H)	Monitor crops and apply CHESS when local pest thresholds are reached and before target populations reach high levels. Continue to monitor crops and make subsequent insecticide applications as necessary. Minimum retreatment interval 14 days. Add a suitable non-ionic surfactant at the recommended rate. Do not apply using equipment carried on the back of the user.
	Suppression of: Silverleaf whitefly (*Bemisia tabaci*)			Monitor crops and apply CHESS when local thresholds for whitefly are reached and before populations reach high levels. Continue to monitor crops and make subsequent insecticide applications as necessary. Minimum retreatment interval 14 days. Target adult whitefly when applying CHESS. DO NOT apply CHESS where whitefly populations predominately consist of eggs, pupae or sessile nymphs unless mixed with another insecticide that has activity against these stages of the whitefly lifecycle. CHESS has no direct activity on immobile stages of the whitefly lifecycle. Add a suitable non-ionic surfactant at the recommended rate. Do not apply using equipment carried on the back of the user.
Tomatoes, eggplants and capsicums (including open field and protected cropping)	Potato aphid (*Macrosiphum euphorbiae),* Green peach aphid (*Myzus persicae*)	200 g/ha	3 days (H)	Monitor crops and apply CHESS when local pest thresholds are reached and before target populations reach high levels. Continue to monitor crops and make subsequent insecticide applications as necessary. Minimum retreatment interval 7 days. When targeting whitefly, apply CHESS to adult populations. DO NOT apply CHESS where whitefly populations predominately consist of eggs, pupae or sessile nymphs unless mixed with another insecticide that has activity against these stages of the whitefly lifecycle. CHESS has no direct activity on immobile stages of the whitefly lifecycle. Add a suitable non-ionic surfactant at the recommended rate.
	Suppression of: Silverleaf whitefly *(Bemisia tabaci),* Greenhouse whitefly *(Trialeurodes vaporariorum)*			
Sweet corn	Corn aphid (*Rhopalosiphum maidis*)	200 g/ha	14 days (H)	Monitor crops and apply CHESS when local pest thresholds are reached and before target populations reach high levels. Continue to monitor crops and make subsequent insecticide applications as necessary. Minimum retreatment interval 14 days. Add a suitable non-ionic surfactant at the recommended rate.

Crop	Pest	Rate	WHP	Critical Comments
Lettuce (all types, including open field and protected cropping)	Brown sowthistle aphid *(Uroleucon sonchi),* Green peach aphid (*Myzus persicae*), Currant lettuce aphid *(Nasonovia ribisnigri)*	200 g/ha	3 days (H)	Monitor crops and apply CHESS when local pest thresholds are reached and before target populations reach high levels. Continue to monitor crops and make subsequent insecticide applications as necessary. Minimum retreatment interval 7 days. When targeting whitefly, apply CHESS to adult populations. DO NOT apply CHESS where whitefly populations predominately consist of eggs, pupae or sessile nymphs unless mixed with another insecticide that has activity against these stages of the whitefly lifecycle. CHESS has no direct activity on immobile stages of the whitefly lifecycle. Add a suitable non-ionic surfactant at the recommended rate. Do not apply using equipment carried on the back of the user.
	Suppression of: Silverleaf whitefly *(Bemisia tabaci)*			
Endive, chicory, radicchio	Currant lettuce aphid *(Nasonovia ribisnigri)*	200 g/ha	3 days (H)	Monitor crops and apply CHESS when local pest thresholds are reached and before target populations reach high levels. Continue to monitor crops and make subsequent insecticide applications as necessary. Minimum retreatment interval 7 days. Add a suitable non-ionic surfactant at the recommended rate. Do not apply using equipment carried on the back of the user.
Leafy vegetables (including spinach, silver beet, kale, endive, mustard, cress, chard, rocket, Asian leafy greens and Chinese cabbage. Including open field and protected cropping)	Potato aphid (*Macrosiphum euphorbiae),* Melon aphid *(Aphis gossypii),* Green peach aphid (*Myzus persicae*)	200 g/ha	3 days (H)	
Cucurbits (including open field and protected cropping)	Melon aphid *(Aphis gossypii),* Green peach aphid (*Myzus persicae*), Potato aphid (*Macrosiphum euphorbiae),* Cowpea aphid *(Aphis craccivora)*	200 g/ha	3 days (H)	Monitor crops and apply CHESS when local pest thresholds are reached and before target populations reach high levels. Continue to monitor crops and make subsequent insecticide applications as necessary. Minimum retreatment interval 7 days. When targeting whitefly, apply CHESS to adult populations. DO NOT apply CHESS where whitefly populations predominately consist of eggs, pupae or sessile nymphs unless mixed with another insecticide that has activity against these stages of the whitefly lifecycle. CHESS has no direct activity on immobile stages of the whitefly lifecycle. Add a suitable non-ionic surfactant at the recommended rate.
	Suppression of: Silverleaf whitefly *(Bemisia tabaci),* Greenhouse whitefly *(Trialeurodes vaporariorum)*			

Crop	Pest	Rate	WHP	Critical Comments
Cotton	Cotton aphid (*Aphis gossypii*)	400 g/ha Add 0.2%v/v organo-silicone surfactant	4 weeks (H)	Apply to a developing aphid population of no greater than 40% infestation and no more than 5 aphids per leaf. Apply prior to row closure on an actively growing crop before honeydew damage has occurred. Thorough spray coverage is essential. CHESS should be used according to the Cotton Industry's Best Management Practices Manual and its associated Spray and Drift Management Plan. CHESS is not a knockdown. DO NOT expect to see a significant reduction in aphid numbers for at least 3 days.
Potatoes	Green Peach Aphid (*Myzus persicae*)	200 g/ha	Nil	Monitor crops and apply CHESS when local pest thresholds are reached and before target populations reach high levels. Continue to monitor crops and make subsequent insecticide applications as necessary. Minimum retreatment interval 14 days.
Stone fruit	Black Peach Aphid (*Brachycaudus persicae),* Green Peach Aphid (*Myzus persicae*)	Dilute spray 20 g / 100L water Concentrate spray Refer to the Application, section	4 weeks (H)	Monitor crops and apply CHESS when local pest thresholds are reached and before target populations reach high levels. Continue to monitor crops and make subsequent insecticide applications as necessary. Minimum retreatment interval 7 days. Apply by dilute or concentrate spraying equipment. Apply the same total amount of product to the target crop whether applying this product by dilute or concentrate spraying methods. Refer to General Instructions for application information. Dilute spraying: If water volumes exceed 2000 L per hectare, total rate of CHESS must not exceed 400 g per hectare. Concentrate spraying: DO NOT use at rates greater than 80 g per 100 L of water.
Almonds and pistachios	Green peach aphid (*Myzus persicae*)	Dilute spray 20 g/100 L Concentrate spray Refer to the Application, section	4 weeks (H)	Monitor crops and apply CHESS when local pest thresholds are reached and before target populations reach high levels. Continue to monitor crops and make subsequent insecticide applications as necessary. Minimum retreatment interval 7 days. Apply by dilute or concentrate spraying equipment. Apply the same total amount of this product to the target crop whether applying via dilute or concentrate spraying methods. Refer to the General Instructions for application information. Dilute spraying: if water volumes exceed 2000 L per hectare, total rate of CHESS must not exceed 400 g product per hectare. Concentrate spraying: DO NOT use at rates greater than 80 g product per 100 L water.
Beetroot	Potato aphid (*Macrosiphum euphorbiae),* Green peach aphid (*Myzus persicae*)	200 g/ha	14 days (H)	Monitor crops and apply CHESS when local pest thresholds are reached and before target populations reach high levels. Continue to monitor crops and make subsequent insecticide applications as necessary. Minimum retreatment interval 14 days. Add a suitable non-ionic surfactant at the recommended rate.

Crop	Pest	Rate	WHP	Critical Comments
Celery	Aphids	200 g/ha	14 days (H)	Apply at early stages of infestation. DO NOT make more than two applications per crop. Minimum re-treatment interval is 14 days. Apply as a foliar spray by ground application equipment only. Use only in field grown crops.
Cut flowers (including Rose, Gerbera and Chrysanthemum) (subject to a tolerance test)	Green peach aphid *(Myzus persicae),* Melon aphid *(Aphis gossypii),* Cowpea aphid *(Aphis craccivora),* Rose aphid *(Macrosiphum rosae)*	Dilute spray 20 g/100 L Concentrate spray Refer to the Application, section	nil	Tolerance Test - CHESS has been successfully used at the recommended doses on a range of species and cultivars without crop damage. However, due to the large number of species and cultivars of cut flowers, cultivar susceptibility MUST always be checked by treating a small number of plants prior to treating the whole crop. Tolerance test conditions, including plant development stage and environmental conditions should be reflective of those expected under subsequent large scale use. Monitor crops and apply CHESS when local pest thresholds are reached and before target populations reach high levels. Continue to monitor crops and make subsequent insecticide applications as necessary. Minimum retreatment interval 14 days. When targeting whitefly, apply CHESS to adult populations. DO NOT apply CHESS where whitefly populations predominately consist of eggs, pupae or sessile nymphs unless mixed with another insecticide that has activity against these stages of the whitefly lifecycle. CHESS has no direct activity on immobile stages of the whitefly lifecycle. Apply by dilute or concentrate spraying equipment. Apply the same total amount of this product to the target crop whether applying via dilute or concentrate spraying methods. Spray in sufficient volume to ensure adequate penetration and coverage. Refer to the General Instructions section of this label for application information. Add a suitable non-ionic surfactant at the recommended rate.
	Suppression of: Silverleaf whitefly *(Bemisia tabaci),* Greenhouse whitefly *(Trialeurodes vaporariorum)*			
Nursery stock (non-food), seedling & plugs, potted colour, trees and shrubs, foliage plants, palms, grasses &	Green peach aphid *(Myzus persicae),* Melon aphid *(Aphis gossypii),* Cowpea aphid *(Aphis craccivora),* Rose aphid *(Macrosiphum rosae)*	Dilute spray 20 g/100 L Concentrate spray Refer to the Application, section	nil	Tolerance Test - CHESS has been successfully used at the recommended doses on a range of species and cultivars without crop damage. However, due to the large number of species and cultivars of nursery stock (non-food), seedling & plugs, potted colour, trees and shrubs, foliage plants, palms, grasses & fruit trees (non-bearing), cultivar susceptibility MUST always be checked by treating a small number of plants prior to treating the whole crop. Tolerance test conditions, including plant development stage and environmental

Crop	Pest	Rate	WHP	Critical Comments
fruit trees (non-bearing) (subject to a tolerance test)	Suppression of: Silverleaf whitefly *(Bemisia tabaci),* Greenhouse whitefly *(Trialeurodes vaporariorum)*			conditions should be reflective of those expected under subsequent large scale use. Monitor crops and apply CHESS when local pest thresholds are reached and before target populations reach high levels. Continue to monitor crops and make subsequent insecticide applications as necessary. Minimum retreatment interval 7 days. When targeting whitefly, apply CHESS to adult populations. DO NOT apply CHESS where whitefly populations predominately consist of eggs, pupae or sessile nymphs unless mixed with another insecticide that has activity against these stages of the whitefly lifecycle. CHESS has no direct activity on immobile stages of the whitefly lifecycle. Apply by dilute or concentrate spraying equipment. Apply the same total amount of this product to the target crop whether applying via dilute or concentrate spraying methods. Spray in sufficient volume to ensure adequate penetration and coverage. Refer to the General Instructions section of this label for application information. Add a suitable non-ionic surfactant at the recommended rate.

H=Harvest

NOT TO BE USED FOR ANY PURPOSE, OR IN ANY MANNER, CONTRARY TO THIS LABEL UNLESS AUTHORISED UNDER APPROPRIATE LEGISLATION.

WITHHOLDING PERIODS

HARVEST WITHHOLDING PERIOD
Fruiting vegetables, Cucurbits, Leafy vegetables, Lettuce: DO NOT HARVEST FOR 3 DAYS AFTER APPLICATION
Brassica vegetables: **DO NOT HARVEST FOR 7 DAYS AFTER APPLICATION**
***Beetroot, sweet corn, celery*: DO NOT HARVEST FOR 14 DAYS AFTER APPLICATION**
Cotton, almonds, pistachios and stonefruit: **DO NOT HARVEST FOR 4 WEEKS AFTER APPLICATION**
Potato: **NOT REQUIRED WHEN USED AS DIRECTED**

GRAZING WITHHOLDING PERIOD
COTTON: DO NOT GRAZE TREATED COTTON CROPS, STUBBLE OR GIN TRASH

Export trade advice
Produce from crops treated with CHESS may contain finite (measurable) residues of pymetrozine and may pose a risk to trade in situations where no residue tolerance (import tolerance) is established in the importing country or where residues in Australian produce are likely to exceed a residue tolerance (import tolerance) established in the importing country. Before you use this product, you are advised to contact Syngenta and/or your industry body about any potential trade issues and their management.

GENERAL INSTRUCTIONS

Mode of Action
CHESS has an anti-feeding effect and is not a knockdown. Users should not expect to see an immediate significant reduction in aphid or whitefly numbers. Although whiteflies and aphids may remain alive for a short time after application of CHESS, further damage to the plant is prevented as feeding is inhibited. Best results are achieved when CHESS is applied when whitefly and aphid populations reach local thresholds and before the target populations reach high levels.

Crop Checking – Cotton
Different methods of scouting will produce different results. The effect of CHESS on aphid populations is best demonstrated by recording the percentage reduction in aphid numbers before and after application, as opposed to recording a decline in the percentage of plants infested. The most reliable scouting method is to visually examine 20 to 30 leaves between the 3rd and 5th highest node of the cotton plant and to estimate or count the number of non-winged aphids present. Scouting for aphids should be made over a random sample of plants, representative of the whole crop area.

Mixing
Add the required amount of CHESS to clean water in a half filled spray tank with the agitator or by-pass in operation. Maintain agitation while filling tank with remainder of water. If a wetter is required, add at completion of filling the tank. Agitation must also be maintained throughout the spray operation.

Application
To be effective CHESS requires thorough spray coverage. Ensure that equipment is properly calibrated to give an even distribution at the correct volume delivering a MEDIUM spray quality.

Cotton and Sweetcorn:
Ground Application: Apply in a sufficient volume of water to ensure thorough coverage
Aerial Application: Apply in a minimum of 30 L water/ha.

Vegetable crops (including brassicas, beetroot, celery, leafy vegetables, tomatoes, capsicums, eggplants, lettuce, endive, radicchio, chicory, cucurbits, and potatoes: Apply via ground rig or knapsack sprayer only in a sufficient volume of water to ensure thorough coverage

Stone fruit, almonds and pistachios, cut flowers and Nursery stock (non-food), seedling & plugs, potted colour, trees and shrubs, foliage plants, palms, grasses & fruit trees (non-bearing): The same quantity of CHESS per hectare should be applied whether spraying either the dilute or concentrate method.
Dilute Spraying: Use a sprayer designed to apply high volumes of water up to the point of runoff and matched to the crop being sprayed. Set up and operate the sprayer to achieve even coverage throughout the crop canopy. Apply sufficient water to cover the crop to the point of runoff. Avoid excessive runoff. The required water volume may be determined by applying different test volumes, using different settings on the sprayer, from industry guidelines or expert advice.
Add the amount of CHESS specified in the Directions for Use table for each 100 L of water. Spray to the point of runoff. If water volumes exceed 2000 L/ha, total rate of CHESS must not exceed 400 g/ha. The required dilute spray volume will change and the sprayer set up and operation may also need to be changed, as the crop grows.
Concentrate Spraying: Use a sprayer designed and set up for concentrate spraying (that is a sprayer which applies water volumes less than those required to reach the point of runoff) and matched to the crop being sprayed. Apply a minimum of 500 L water/ha. Set up and operate the sprayer to achieve even coverage throughout the crop canopy using your chosen water volume. Determine an appropriate dilute spray volume (see Dilute Spraying) for the crop canopy. This is needed to calculate the concentrate mixing rate. The mixing rate for concentrate spraying can then be calculated in the following way:

Example only
1. Dilute spray volume as determined above: for example 2000 L/ha
2. Your chosen concentrate spray volume: for example 500 L/ha
3. The concentrate factor in this example is 4X (ie 2000 L ÷ 500 L = 4)
4. If the dilute label rate is 20 g/100 L, then the concentrate rate becomes 4 x 20, that is 80 g/100 L of concentrate spray.

The chosen spray volume, amount of product per 100 L of water, and the sprayer set up and operation may need to be changed as the crop grows. For further information on concentrate spraying, users are advised to consult relevant industry guidelines, undertake appropriate competency training and follow industry Best Practices.

Compatibility
CHESS is physically compatible with AFFIRM®, VERTIMEC® and KARATE® ZEON®. As formulations of other manufacturers' products are beyond the control of Syngenta Australia Pty Ltd, and water quality varies with location, all mixtures should be tested prior to mixing commercial quantities.

Insecticide Resistance Warning

GROUP	9B	INSECTICIDE

For insecticide resistance management, CHESS Insecticide is a Group 9B insecticide. Some naturally occurring insect biotypes resistant to CHESS and other Group 9B insecticides may exist through normal genetic variability in any insect population. The resistant individuals can eventually dominate the insect population if CHESS and other Group 9B insecticides are used repeatedly. The effectiveness of CHESS on resistant individuals could be significantly reduced. Since occurrence of resistant individuals is difficult to detect prior to use, Syngenta Australia Pty Ltd Crop Protection accepts no liability for any losses that may result from failure of CHESS to control resistant insects.
CHESS may be subject to specific resistance management strategies. For further information contact your supplier, Syngenta representative or local agricultural department agronomist.

PRECAUTION
DO NOT use human flaggers/markers unless they are protected by engineering controls such as enclosed cabs.

Re-entry Period: DO NOT allow entry into treated areas until the spray has dried. If prior entry is necessary, wear cotton overalls buttoned to the neck and wrist (or equivalent clothing) and chemical resistant gloves. Clothing must be laundered after each day's use.

PROTECTION OF WILDLIFE, FISH, CRUSTACEANS AND ENVIRONMENT
May be hazardous to aquatic invertebrates. DO NOT contaminate streams, rivers or waterways with the chemical or used container. DO NOT apply under meteorological conditions or from spraying equipment which could be expected to cause spray to drift onto adjacent areas, particularly wetlands, waterbodies or watercourses. DO NOT spray across open bodies of water.

STORAGE AND DISPOSAL
Store in the closed, original container in a dry, cool, well-ventilated area out of direct sunlight. Shake empty container into spray tank. Single rinse before disposal. Add rinsings to spray tank. DO NOT dispose of undiluted chemicals on site. Puncture or shred and bury empty containers in a local authority landfill. If no landfill is available, bury the containers below 500 mm in a disposal pit specifically marked and set up for this purpose clear of waterways, desirable vegetation and tree roots. Empty containers and product should not be burnt.

SAFETY DIRECTIONS
Will irritate the eyes and skin. Avoid contact with eyes and skin. Wash hands after use. When opening the container, preparing the spray and using* the prepared spray wear:
- **cotton overalls buttoned to the neck and wrist (or equivalent clothing)**
- **a washable hat**
- **elbow-length chemical resistant gloves**

Except when preparing the spray for boomspray equipment wear cotton overalls, over normal clothing, buttoned to the neck and wrist.
In addition if applying by low pressure handwand wear elbow-length chemical resistant gloves and disposable dust/mist face mask covering mouth and nose.
When preparing the spray for equipment carried on the back of the user, wear:
- cotton overalls buttoned to the neck and wrist
- elbow-length chemical resistant gloves.
If applying by equipment carried on the back of the user, wear cotton overalls over normal clothing buttoned to the neck and wrist.
Wash hands after use.
After each day's use, wash gloves and contaminated clothing.
*** Pilots of aircraft are not required to wear protective equipment during application.**

FIRST AID
If poisoning occurs contact a doctor or Poisons Information Centre. Phone 131 126.

SAFETY DATA SHEET
If additional hazard information is required refer to the Safety Data Sheet. For a copy telephone 1800 067 108 or visit our website at www.syngenta.com.au

Suspected of causing cancer

DISCLAIMER:
This product complies with the specifications in its statutory registration. Implied terms and warranties are excluded. Syngenta's liability for breach of the express or any non-excludable implied warranty is limited to product replacement or purchase price refund. The purchaser must determine suitability for intended purpose and take all proper precautions in the handling, storage and use of the product including those on the label and/or safety data sheet failing which Syngenta shall have no liability.

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Batch No.	
Date of Manufacture	